IAT New Test 3

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 An alga has phycoerythrin, stores floridean starch and lacks motile reproductive cells. Which class is supported by all three observations?
A) Chlorophyceae
B) Phaeophyceae
C) Rhodophyceae
D) Bryophyta

Q2 In a young root, radial xylem and phloem lie inside a cell layer whose radial walls contain a suberin-rich Casparian strip. What does that layer chiefly regulate?
A) Apoplastic entry into the stele by forcing selective membrane crossing
B) Photosynthetic gas exchange in mesophyll
C) Longitudinal phloem pressure from leaf to sink
D) Direct pollen transfer to a stigma

Q3 A carrier imports Na^+ and glucose together even when cellular glucose is higher than outside. Blocking the ATP-driven Na^+ pump eventually stops glucose entry, although the carrier remains present. What powers uptake?
A) Glucose's own downhill gradient
B) Direct ATP hydrolysis at the glucose carrier is established by the result
C) Simple diffusion through membrane lipid
D) The Na^+ gradient maintained by the pump drives coupled glucose transport

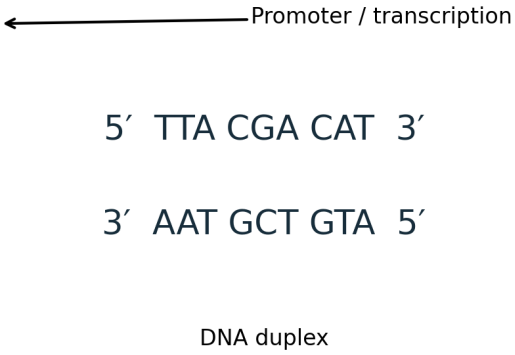
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Q4 Identical illuminated leaves have equal stomatal aperture and leaf temperature. X is in humid air and Y in dry air, with equal air movement. Initially, which loses more water vapour and why?
A) X, because humid air pulls vapour more strongly
B) Y, because its outward vapour-pressure difference is larger
C) Both equally, because aperture alone sets diffusion
D) Neither, because transpiration requires darkness

Q5 Two tissues receive blood of the same arterial oxygen content. Tissue Q produces more CO_2 and has lower local pH than P. What haemoglobin response helps deliver oxygen to Q?
A) Higher pH at Q prevents release
B) Haemoglobin loses all binding capacity permanently
C) Lower pH favours unloading at a given oxygen partial pressure
D) CO_2 creates additional oxygen in blood

Q6 Bagged flowers of P set no seed after hand self-pollination, but set seed after compatible cross-pollination. P's pollen works on another compatible plant. What inference is supported by these controls?
A) Failure is specific to the self combination, consistent with self-incompatibility
B) All P pollen is universally nonviable
C) P ovules never develop following fertilisation
D) Bagging always prevents seed formation

Q7 The figure shows a short DNA duplex and a promoter arrow pointing right to left. Assume transcription starts at the rightmost shown base, uses the strand read 3′→5′, and ends after the shown sequence. Which mRNA is made, written 5′→3′?



A) 5′-UUA CGA CAU-3′
B) 5′-AAU GCU GUA-3′
C) 5′-UAC AGC AUU-3′
D) 5′-AUG UCG UAA-3′

Q8 A person loses many helper T lymphocytes while neutrophil numbers initially remain normal. Which impairment best follows from the helper-cell role?
A) Only clot formation is lost
B) Coordination of both antibody and cell-mediated adaptive responses weakens
C) All pathogens immediately stop dividing
D) Innate neutrophils become the sole source of antigen-specific antibodies

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Q9 PCR tubes X and Y have the same template, polymerase, nucleotides and cycling. X has both primers flanking a target; Y lacks the reverse primer. What follows for the defined double-stranded target after many ideal cycles?
A) Y must amplify the bounded target more rapidly
B) Both amplify exponentially at the same rate
C) X can amplify it exponentially; Y cannot sustain that bounded two-primer process
D) Neither can extend any DNA strand

Q10 A plant community has 1000 units net production in an interval. Herbivores consume 250 units and 100 units remain as unconsumed plant biomass; all the rest enters detritus. How many units enter detritus?
A) 650
B) 750
C) 900
D) 1150

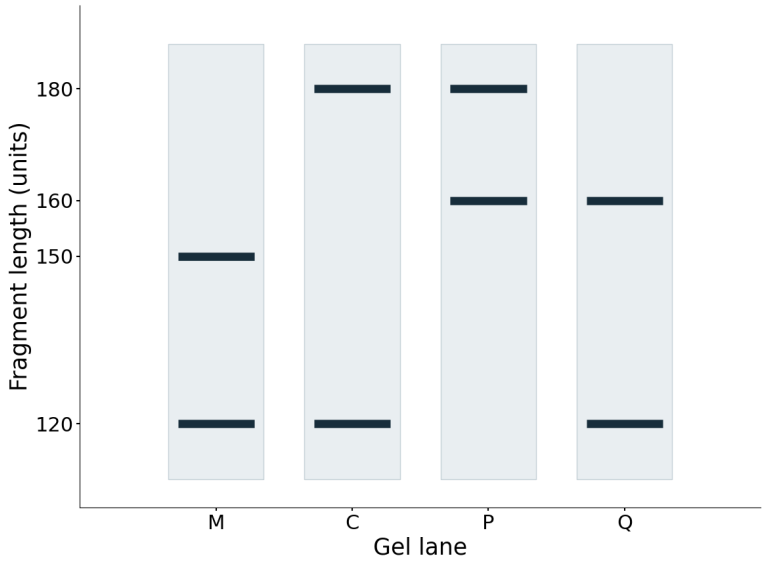
Q11 Pigment needs two independent functional enzymes encoded by A and B; aa or bb blocks the pathway. Two AaBb plants cross. What fraction of offspring is pigmented under complete dominance?
A) 3/4
B) 1/4
C) 7/16
D) 9/16

Q12 In strain P, a functional repressor cannot bind its mutated operator; promoter and structural genes work. Strain Q has a normal operator and repressor. With no glucose, compare enzyme expression without and with lactose.
A) Both express only without lactose
B) P expresses in both; Q is low without lactose and induced with it
C) P is always off; Q is always on
D) Both remain off because a promoter is present

Q13 Presynaptic stimulation releases normal transmitter in two preparations. A postsynaptic response occurs in P but not Q; direct electrical stimulation of Q's postsynaptic membrane still gives a normal action potential. Where is the likely defect?
A) Presynaptic transmitter synthesis
B) Axonal action-potential generation in Q's postsynaptic membrane
C) Postsynaptic transmitter reception or coupling
D) Complete absence of the presynaptic neuron

Q14 In ordinary meiosis, when does chromosome DNA replication occur relative to meiosis I and II?
A) Once before meiosis I, without replication between the divisions
B) Once before each division
C) Only after meiosis II
D) Never, because chromosomes are already visible

Q15 The figure shows one inherited marker in mother M, child C and candidates P and Q. Assume one allele from each biological parent, precise band sizing and no mutation. What follows from this marker alone?



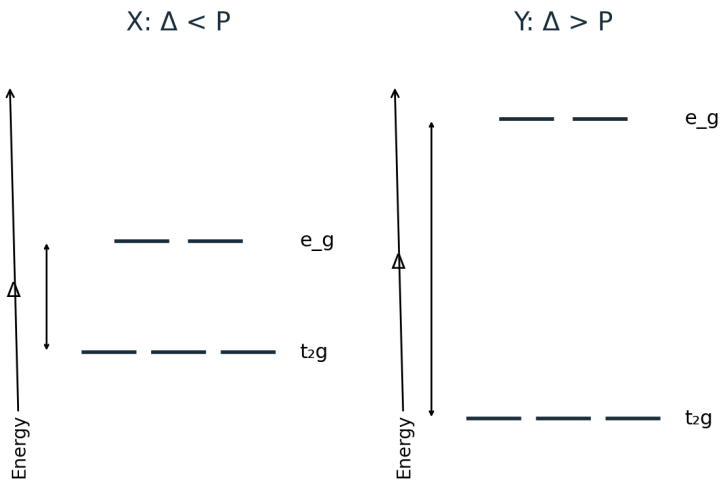
A) Q is established as father with certainty
B) P is compatible and Q excluded, but P is not uniquely proved by one marker
C) Both are excluded because C shares a band with M
D) Each candidate is certainly a parent because each has two bands

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Chemistry

Q16 Each C–O bond in CO_2 is polar. Under its linear geometry, what is the molecule's net dipole moment?
A) Equal to one bond dipole
B) Zero because equal opposite bond dipoles cancel
C) Twice one bond dipole along the same direction
D) Undefined because carbon is central

Q17 The figure shows small octahedral splitting for complex X and large splitting for Y, both d^6 at the same metal oxidation state. Under the simple crystal-field model, how many unpaired electrons do X and Y have?



A) 0 and 4
B) 2 and 2
C) 6 and 0
D) 4 and 0

Q18 Between Mn^{2+} (Z=25) and Fe^{2+} (Z=26), which has a half-filled 3d subshell and how many 3d electrons does it contain?
A) Mn^{2+}, five
B) Fe^{2+}, five
C) Mn^{2+}, seven
D) Fe^{2+}, six

Q19 Why is first ionisation energy of N higher than O, despite O's larger nuclear charge?
A) Nitrogen has a complete $2p^6$ subshell
B) Oxygen has no 2p electrons
C) Oxygen has a paired 2p orbital with extra repulsion relative to N's $2p^3$ arrangement
D) Nitrogen's outer electrons occupy n=3

Q20 A proton of mass m, charge e and alpha particle of mass 4m, charge 2e start from rest and traverse the same accelerating potential V nonrelativistically. Find $\lambda_{proton}/\lambda_{\alpha}$.
A) 2
B) $\sqrt{8}$
C) $\sqrt{2}$
D) 8

Q21 A+B→C uses 0.10 mol A and 0.16 mol B in a 1:1 reaction, isolating C at 80% theoretical yield. Then 2C→D isolates D at 75% of the theoretical yield from the isolated C. How many moles D result?
A) 0.060
B) 0.040
C) 0.075
D) 0.030

Q22 At the para position, compare nitro- and methoxy-substituted benzoic acids with benzoic acid under the same conditions. Which is expected to be strongest?
A) p-Nitrobenzoic acid, because the electron-withdrawing group stabilises benzoate
B) p-Methoxybenzoic acid, because resonance donation stabilises benzoate
C) Benzoic acid, because substitution can never affect acidity
D) All exactly equal, because each has a COOH group

Q23 The figure shows the two distinct carbonyl products from reductive ozonolysis of one acyclic alkene molecule, one molecule of each. Which alkene reconstructs the original carbon skeleton?

Reductive ozonolysis products

$$H\text{–}C(=O)\text{–}H \quad + \quad CH_3\text{–}C(=O)\text{–}H$$

one molecule of each

A) Ethene
B) But-2-ene
C) Propene
D) 2-methylpropene

Q24 Under mild conditions that permit ordinary SN2 substitution of a primary bromoalkane by hydroxide, bromobenzene is much less reactive. Which structural distinction matters most?
A) Bromobenzene has no C–Br bond
B) Resonance gives aryl C–Br partial double-bond character and sp^2 carbon disfavors backside SN2 attack
C) A primary bromoalkane has a benzene ring
D) Bromobenzene is freely ionised into phenyl cations in water

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Q25 Phenol forms a salt with ordinary aqueous NaOH, whereas ethanol is not comparably deprotonated. What explains the difference?
A) Ethanol contains no O–H bond
B) Phenol is an aldehyde
C) Hydroxide reacts only with aromatic C–C bonds
D) Phenoxide is resonance-stabilised relative to an alkoxide

Q26 Complete hydrolysis of Gly–Ala yields free glycine and alanine. At pH well above both isoelectric points, what net-charge tendency does each free amino acid have?
A) Both tend net negative
B) Both tend net positive
C) Glycine positive, alanine negative solely due to side chains
D) Both must be net neutral at every pH

Q27 AgCl has Ksp=1.0×10^{-10} at the stated temperature. Approximate equilibrium [Ag^+] in 0.10 M NaCl if its dissolution changes [Cl^-] negligibly.
A) 1.0×10^{-5} M
B) 1.0×10^{-10} M
C) 1.0×10^{-9} M
D) 0.10 M

Q28 At fixed temperature and composition, a liquid mixture's total vapour pressure exceeds its ideal Raoult-law prediction. Which interaction pattern is consistent with this positive deviation?
A) Unlike A–B attractions must be stronger than like pairs
B) Unlike A–B attractions are weaker on average, permitting easier escape
C) The mixture has zero vapour pressure
D) All molecular attractions vanish

Q29 Aqueous $CuSO_4$ is electrolysed with inert Pt electrodes under conditions where Cu^{2+} deposits at the cathode and water is oxidised at the anode. Which primary products follow?
A) H_2 and Cu
B) Cu and SO_2
C) Cu and gaseous sulfate
D) Cu and O_2

Q30 50.0 mL of 1.00 M monoprotic acid and 50.0 mL of 0.500 M strong base react fully in an insulated calorimeter. The final 100 g solution warms 6.0 °C; use c=4.20 J g^{-1} K^{-1} and neglect calorimeter heat. Assuming only neutralisation causes the measured heat, find enthalpy per mole water formed.
A) −100.8 kJ/mol
B) −50.4 kJ/mol
C) +100.8 kJ/mol
D) −201.6 kJ/mol

Mathematics

Q31 The circle $x^2+y^2-6x-8y=0$ is touched by a line $3x+4y=k$. What are the possible values of k?
A) -25 and 25
B) -5 and 45
C) 25 only
D) 0 and 50

Q32 The point (1,1) lies on $x^2+xy+y^2=3$. Where does its tangent meet the x-axis?
A) $x=-2$
B) $x=1$
C) $x=2$
D) It is parallel to the x-axis

Q33 Equal squares of side x are cut from the four corners of a 12 cm by 8 cm rectangle, and the sides are folded into an open box. Which x gives maximum volume?
A) $(10+2\sqrt{7})/3$ cm
B) $(10-2\sqrt{7})/3$ cm
C) 2 cm
D) 4 cm

Q34 Evaluate $\int_0^1[x/(1+x^2)-x/(1+x)]dx$.
A) $(3/2)\ln 2-1$
B) $1-(3/2)\ln 2$
C) $(1/2)\ln 2$
D) $\ln 2-1$

Q35 Let $f(x)=\sqrt{(x-1)}$ and $g(t)=1/(t-2)$, both real-valued where defined. What is the domain of $g\circ f$?
A) $x>1$ only
B) $x\geq 5$ only
C) All real x except 5
D) $x\geq 1$ with $x\neq 5$

Q36 Vectors $a=(1,1,0)$, $b=(1,0,1)$, $c=(t,1,1)$ span a parallelepiped of volume 3. What real t-values are possible?
A) 2 only
B) -5 and 1
C) -1 and 5
D) 1 and 5

Q37 A has rows R_1,R_2,R_3 and $\det A=3$. Matrix B has rows R_1+R_2, R_2+R_3, R_3+R_1 in that order; then its first and third rows are interchanged. Find $\det B$.
A) $+6$
B) -6
C) -3
D) $+3$

Q38 Three balls are drawn together from a bag with 4 red and 3 blue balls. Given that at least two drawn balls are red, what is the probability that all three are red?
A) 2/11
B) 4/35
C) 1/2
D) 3/11

Q39 A path from (0,0) to (5,4) uses only unit right or up steps. How many such paths avoid the point (2,2)?
A) 126
B) 60
C) 36
D) 66

Q40 For integer $n \geq 4$, the coefficient of x^3 in $(1+x)^n(1-x)$ vanishes. What is the coefficient of x^4 in the same polynomial?
A) $+5$
B) -10
C) -5
D) 0

Q41 What is $\Sigma_{k=1}^{4}\ 1/[k(k+2)]$?
A) 11/24
B) 17/30
C) 21/40
D) 4/5

Q42 For which real k does $x^2-(k+1)x+(k-2)=0$ have one negative root and one positive root strictly greater than 2?
A) $0<k<2$
B) $k<2$
C) $0 \leq k \leq 2$
D) $k>2$

Q43 Five observations have arithmetic mean 4 and sum of squares 125. What is their population standard deviation?
A) 9
B) 45
C) $\sqrt{45}$
D) 3

Q44 Which differential equation is satisfied by every function $y=C_1e^x+C_2e^{(-x)}$, for arbitrary constants C_1, C_2?
A) $y' = y$
B) $y'' = -y$
C) $y'' = y$
D) $y' = 0$

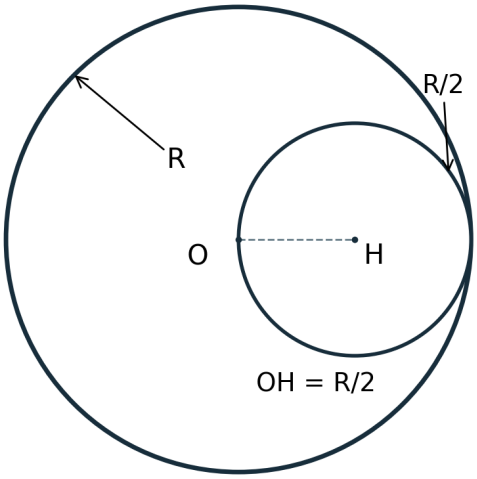
Q45 Define $f(x)=\sin(2x)/x$ for $x \neq 0$ and $f(0)=k$. For continuity at zero, what are k and $f'(0)$?
A) $k=0,\ f'(0)=2$
B) $k=2,\ f'(0)=0$
C) $k=2,\ f'(0)=2$
D) No value of k works

Physics

Q46 From the same height, one ball is released horizontally and another is dropped from rest at the same instant. Neglect air resistance. Which lands first on level ground?
A) Both simultaneously
B) Horizontally released ball
C) Dropped ball
D) It depends on horizontal speed

Q47 At an instant, $v=(3\hat{i}+4\hat{j})$ m/s and force $F=(2\hat{i}-\hat{j})$ N. What instantaneous power does this force supply?
A) 10 W
B) 5 W
C) −2 W
D) 2 W

Q48 The figure shows a uniform disc of original mass M and radius R with a circular hole of radius R/2 removed; its centre is R/2 from original centre O. About the perpendicular axis through O, what is the remaining moment of inertia?



A) $3MR^2/8$
B) $15MR^2/32$
C) $13MR^2/32$
D) $7MR^2/16$

Q49 An open water tank accelerates to the right at g/2. In its frame, what is the magnitude of the steady free-surface slope, vertical rise divided by horizontal distance?
A) 2
B) 1/2
C) 1
D) 1/4

Q50 Two identical bodies of constant heat capacity C start at absolute temperatures T and 3T. Isolated together, they equilibrate without work. What is their total entropy change?
A) C ln(4/3)
B) 0
C) C ln(3/4)
D) 2C ln2

Q51 At a fixed instant, two points along a sinusoidal travelling wave are separated by λ/6 in the propagation direction. What is their phase difference modulo 2π?
A) π/6
B) π/2
C) 2π/3
D) π/3

Q52 A satellite of mass m orbits a planet of mass M in a circle of radius r. Taking potential energy zero at infinity, what is its total mechanical energy?
A) +GMm/(2r)
B) −GMm/r
C) −GMm/(2r)
D) 0

Q53 The figure places +q at x=0 and −4q at x=3a, with a>0. At which finite points on the x-axis is the electric potential zero?



A) x=a and x=−3a
B) x=−a and x=3a/5
C) x=0 and x=3a
D) x=−3a/5 and x=a

Q54 A parallel-plate capacitor stays connected to an ideal constant-voltage battery while plate separation is doubled, with area and dielectric unchanged. How do stored charge and energy change?
A) Both become half
B) Both double
C) Charge halves but energy doubles
D) Charge stays fixed and energy halves

Q55 A cell of emf 12 V and internal resistance 2 Ω supplies a 4 Ω external resistor. What is the terminal voltage while it delivers current?
A) 12 V
B) 4 V
C) 6 V
D) 8 V

Q56 The plane of a single-turn loop of area A carrying current I makes 30° with a uniform magnetic field B. What is the torque magnitude?
A) IAB/2
B) IAB
C) (√3/2)IAB
D) Zero

Q57 An ideal transformer has 200 primary turns and 50 secondary turns. Its primary is at 240 V rms; the secondary drives a 6 Ω resistive load. What are secondary voltage and primary current?
A) 60 V and 10 A
B) 60 V and 2.5 A
C) 240 V and 2.5 A
D) 120 V and 5 A

Q58 A ray crosses a slab of thickness L normally, refractive index n. Relative to the same geometric distance in vacuum, what is excess optical path length?
A) (n−1)L
B) nL
C) L/n
D) (n+1)L

Q59 An electromagnetic plane wave has electric field along +y and magnetic field along +z. What direction does it propagate?
A) −x
B) +y
C) +z
D) +x

Q60 Equal initial numbers N_0 of nuclides X and Y have half-lives 2 h and 4 h. After 8 h, find (i) activity ratio AX:AY and (ii) fraction of all initial nuclei still undecayed.
A) 1:4 and 5/32
B) 2:1 and 5/32
C) 1:2 and 5/32
D) 1:2 and 27/32

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / C Algal pigments and reserve food

Answer: (C) Rhodophyceae.

Why: Red algae commonly possess phycoerythrin and store floridean starch. The lack of flagellated reproductive stages supplies a third consistent clue, so classification does not rest on visible colour alone. These three independent traits are more diagnostic than apparent colour alone, which can vary with lighting or accessory pigments. Their combination also excludes a land-plant life cycle.

Why not the others: (A) Green algae characteristically store starch and chlorophyll predominates. (B) Brown algae have fucoxanthin and laminarin/mannitol reserves. (D) Bryophytes are embryophyte land plants, not this algal class.

Key point: Combine pigments, storage material and life-cycle characters.

Q2 / A Root endodermis and Casparian strip

Answer: (A) Endodermal selective entry.

Why: Water moving through cell walls encounters the waterproof strip at the endodermis. To pass inward toward the stele, it must cross a living membrane, where transport can be regulated. The radial vascular arrangement and wall band identify the location. The strip blocks the apoplastic wall route at a continuous endodermal boundary. Water and solutes must enter a symplast through selective membranes before reaching the vascular stele.

Why not the others: (B) Mesophyll is in leaves. (C) Phloem transport occurs after vascular entry and is not the strip's role. (D) Is reproductive transport unrelated to roots.

Key point: Relate wall modification to the path water must take.

Q3 / D Secondary active transport

Answer: (D) Coupling to the Na^+ gradient.

Why: Glucose travels uphill, so passive diffusion cannot explain it. The pump uses ATP to maintain a downhill Na^+ gradient; cotransport can harness that gradient. Blocking the pump removes the stored gradient over time even while the cotransporter protein remains. The Na^+ gradient provides free energy for cotransport, while ATP is consumed indirectly by the pump that maintains that gradient. Blocking the pump eventually collapses the driver even if the carrier itself still exists.

Why not the others: (A) Glucose is stipulated uphill. (B) The data do not show ATP acting directly on this carrier. (C) Lipid diffusion neither couples Na^+ nor sustains uphill glucose movement.

Key point: Distinguish direct ATP use from gradient-mediated secondary transport.

Q4 / B Vapour-pressure gradient and transpiration

Answer: (B) Y in dry air.

Why: Water vapour is near saturation in internal leaf spaces. Lower external humidity creates a larger concentration or vapour-pressure gradient through otherwise similar stomata, increasing outward diffusion initially. Holding temperature, aperture and movement constant isolates humidity. Transpiration follows the vapour-pressure gradient from moist internal spaces to outside air. Equal stomatal aperture holds pore conductance approximately fixed, isolating the external humidity difference.

Why not the others: (A) Humidity reduces the gradient. (C) Equal aperture does not eliminate a changed driving difference. (D) Light is present and darkness is not required.

Key point: Flux depends on both conductance and driving gradient.

Q5 / C Bohr effect in active tissue

Answer: (C) Enhanced oxygen unloading.

Why: More CO_2 and acidity shift haemoglobin's oxygen dissociation relation toward lower affinity. At the same local oxygen pressure, more bound oxygen can therefore be released to metabolically active Q. This is a reversible local response, not destruction of haemoglobin. At the same pO_2 a right-shifted curve has lower saturation, so more oxygen can leave the blood in metabolically active tissue. The higher CO_2 is a local consequence of respiration, not a change in arterial delivery.

Why not the others: (A) Reverses the stated pH change. (B) Falsely makes modulation permanent. (D) CO_2 does not synthesize oxygen.

Key point: Relate metabolic CO_2 production to oxygen delivery.

Q6 / A Self-incompatibility controls

Answer: (A) Self-combination rejection.

Why: Cross pollen produces seed on P, establishing that its ovules and bagging procedure can support reproduction. P pollen succeeds elsewhere, establishing viability. The remaining contrast is the self combination, consistent with a self-incompatibility response. The cross-pollination control shows the stigma, ovules and bagging method can produce seeds. Failure only with self pollen supports a self-recognition barrier rather than sterility or accidental damage.

Why not the others: (B) Contradicts successful P pollen use. (C) Contradicts cross seed set. (D) Bagged hand crossing worked.

Key point: Reciprocal controls isolate partner specificity from general infertility.

Q7 / D Transcription direction and antiparallel DNA strands

Answer: (D) 5′-AUG UCG UAA-3′.

Why: Moving right to left, polymerase must read a 3′→5′ template, which is the top strand in this orientation. Reverse its base order to TAC AGC ATT, then make the complementary RNA AUG UCG UAA. Equivalently, read the bottom coding strand right to left and replace T by U. Both direction and complementarity matter. The top strand is read in its 3′→5′ direction only when moving from right to left. Complementary RNA is synthesized 5′→3′; reverse the written order before replacing template bases with RNA complements.

Why not the others: (A) Copies the displayed top strand in the wrong direction. (B) Copies the bottom strand left to right. (C) reverses without making the RNA complement.

Key point: Select template by promoter direction before transcribing antiparallel bases.

Q8 / B Helper T-cell coordination

Answer: (B) Broad adaptive coordination weakens.

Why: Helper T cells signal to other immune cells, including B cells and cytotoxic T cells. Their sustained loss can thus impair both antibody production and cell-mediated responses even when a neutrophil count is initially normal. A count of one innate cell type does not replace adaptive coordination. Helper T-cell signals coordinate antibody production and cellular immunity. Normal neutrophil count does not restore antigen-specific B-cell help, so broad adaptive responses become weaker.

Why not the others: (A) Clotting is not the central helper-cell function. (C) No immune loss sterilises every pathogen. (D) Neutrophils do not secrete antigen-specific antibody.

Key point: Trace effects through the cells a helper population coordinates.

Q9 / C Two-primer PCR amplification

Answer: (C) Exponential amplification requires both primers.

Why: Opposed primers define both ends and allow newly synthesized target strands to serve as templates on subsequent cycles. With one missing primer, extension from the remaining primer may occur, but the same bounded double-stranded product cannot repeatedly generate itself by two-primer cycling. Each primer binds an opposite template strand with a 3′ end directed into the target interval. One missing primer permits only linear copying of one side, whereas two define products that amplify exponentially.

Why not the others: (A) Missing a primer does not improve target amplification. (B) Overlooks the second defined end. (D) One primer can still support some extension.

Key point: Distinguish any extension from exponential amplification of a defined amplicon.

Q10 / A Energy partition into detritus

Answer: (A) 650 units.

Given: The prompt partitions net production into consumption, retained biomass and detritus.

Formula/Principle: Conservation under the stated exhaustive partition.

Step 1: Accounted elsewhere=250+100=350 units.

Step 2: Detritus=1000−350=650 units.

Why: Detritus is not identical to every unconsumed unit because some plant material remains stored during the interval. Net production is already after producer respiration. The stem explicitly partitions it into herbivore consumption, retained biomass and detritus, so subtract exactly the first two non-detrital flows.

Why not the others: (B) ignores stored biomass; (C) ignores herbivory; (D) adds a flow to net production.

Key point: Check whether categories overlap before subtracting.

Q11 / D Complementary gene action

Answer: (D) 9/16.

Given: Pigment requires A_ and B_ together.

Formula/Principle: Independent segregation allows multiplication of the two dominant-phenotype probabilities.

Step 1: Each Aa×Aa locus gives functional phenotype probability 3/4.

Step 2: Both present=$(3/4)^2$=9/16.

Why: Either homozygous recessive block prevents the final pigment even if the other enzyme is present. The pathway is sequential: product from the A-controlled step is substrate for the B-controlled step. A functioning enzyme at only one step cannot bypass the missing other step.

Why not the others: (A) checks only one locus; (B) treats two recessive classes as the requirement; (C) is the white fraction.

Key point: Follow the whole sequential pathway, not one gene alone.

Q12 / B Lac operon operator defect

Answer: (B) P constitutive, Q inducible.

Why: In P the repressor cannot occupy its DNA site even without inducer, so transcription is not blocked by this control. In Q, bound repressor suppresses transcription until lactose-derived inducer reduces its binding. With glucose absent, the comparison isolates the operator–repressor response rather than catabolite effects. The operator is a cis-acting DNA site, whereas the repressor protein acts in trans. An operator mutation can make a functional repressor unable to block that operon's structural-gene transcription.

Why not the others: (A) Reverses induction. (C) Reverses both strain behaviours. (D) A working promoter permits transcription rather than forbidding it.

Key point: Separate a defective DNA binding site from a defective protein.

Q13 / C Synaptic receptor versus membrane excitability

Answer: (C) Postsynaptic reception or coupling.

Why: Normal measured release argues against presynaptic failure. Direct electrical stimulation shows that the postsynaptic membrane can still generate an action potential. The missing response lies between arrival of transmitter and electrical activation, such as receptor binding or its coupling to channels. The presynaptic transmitter measurement rules out a failure of release. A response to direct stimulation confirms downstream excitability, narrowing the lesion to transmitter recognition or the postsynaptic receptor process.

Why not the others: (A) and (D) contradict normal transmitter release. (B) contradicts successful direct electrical activation.

Key point: Use a bypass experiment to localise a failure in a signalling chain.

Q14 / A Replication before the two meiotic divisions

Answer: (A) Replication precedes meiosis I only.

Why: S phase copies each chromosome into sister chromatids before homologues separate in the first division. Meiosis II separates those sisters without a second intervening DNA-replication round, allowing haploid products. DNA synthesis occurs once during the preceding interphase. Meiosis I separates homologous chromosomes, and meiosis II separates sister chromatids without another intervening S phase.

Why not the others: (B) Would duplicate DNA again between divisions. (C) Comes too late for segregation. (D) Visible chromosomes have already been replicated.

Key point: Distinguish one replication from two consecutive divisions.

Q15 / B DNA marker parentage exclusion

Answer: (B) P compatible, Q excluded.

Why: The child can inherit 120 from M; the 180 allele must then come from the other biological parent. P has 180 and Q does not, so Q is excluded under the assumptions. Many untested people might carry 180, so a single marker cannot establish P uniquely. Every child band must be assigned to a parent under the stated single-marker assumptions. The 180 band cannot be supplied by M, so compare that band, rather than overall resemblance, with candidate lanes.

Why not the others: (A) Q lacks 180. (C) Sharing a maternal allele is expected. (D) Two bands alone show neither transmission nor identity.

Key point: A marker may exclude a candidate without proving another uniquely.

Chemistry

Q16 / B Bond dipoles and molecular geometry

Answer: (B) Zero.

Why: The two identical bond dipole vectors lie on the same line but point in opposite directions from the central carbon. Vector addition cancels them even though each individual C–O bond remains polar. Molecular polarity depends on geometry as well as bond polarity. Each C–O bond contributes an equal dipole in an opposite direction along the O=C=O axis. Vector addition gives $\boldsymbol{\mu}_{\text{net}} = \boldsymbol{\mu}_1 + \boldsymbol{\mu}_2 = 0$, even though individual bonds are polar.

Why not the others: (A) Keeps only one bond. (C) Adds magnitudes without directions. (D) A central atom does not prevent vector addition.

Key point: Sum bond dipoles as vectors.

Q17 / D High- and low-spin octahedral d6

Answer: (D) Four and zero.

Why: The small splitting in X makes occupation of the upper eg levels preferable to pairing too early: high-spin d^6 is $t_2g^4e_g^2$, with four unpaired electrons. The large gap in Y favours pairing in t_2g first: $t_2g^6e_g^0$ has no unpaired electrons. The figure supplies the energetic comparison but leaves the filling to be reasoned out. For d^6, the high-spin small-gap arrangement leaves more electrons unpaired in both lower and upper sets; the large-gap case pairs in t_2g first. Compare crystal-field energy cost with pairing energy, not merely total d-electron count.

Why not the others: (A) Reverses complexes. (B) Counts only eg electrons in X and invents unpaired electrons in Y. (C) Treats all six d electrons as independently unpaired despite only five orbitals.

Key point: Compare Δ with pairing energy before filling orbitals.

Q18 / A Half-filled d subshell in common ions

Answer: (A) Mn^{2+}, $3d^5$.

Why: Neutral Mn is $[Ar]3d^54s^2$ and neutral Fe is $[Ar]3d^64s^2$. Forming 2+ ions removes the two 4s electrons first, leaving Mn^{2+} $3d^5$ and Fe^{2+} $3d^6$. Five d electrons give a half-filled five-orbital subshell. The 4s electrons are removed before 3d electrons for these cations. Thus Mn^{2+} remains $3d^5$ while Fe^{2+} becomes $3d^6$; only the former has the half-filled d subshell.

Why not the others: (B) Assigns Mn's count to Fe. (C) Retains 4s electrons as 3d. (D) Correctly counts Fe but six is not half-filled.

Key point: Remove ns electrons before (n−1)d in these cations.

Q19 / C Nitrogen–oxygen ionisation anomaly

Answer: (C) Pairing repulsion in O.

Why: N has one electron in each of the three 2p orbitals, a relatively stable half-filled arrangement. O adds a fourth 2p electron that pairs in one orbital; repulsion makes removal of one electron easier than a simple nuclear-charge trend suggests. Oxygen's fourth 2p electron shares one orbital and experiences extra electron–electron repulsion. Removing it relieves a pairing cost, whereas removing from nitrogen disrupts a stable singly occupied pattern.

Why not the others: (A) N is $2p^3$, not $2p^6$. (B) O is $2p^4$. (D) Both valence shells are n=2.

Key point: Consider orbital occupancy when a periodic trend has an exception.

Q20 / B de Broglie wavelengths after acceleration

Answer: (B) √8.

Given: Kinetic energy of each is qV; alpha mass and charge factors are 4 and 2.

Formula/Principle: λ=h/√(2mqV).

Step 1: pα²/pp²=(4m×2e)/(m×e)=8.

Step 2: Since λ is inverse momentum, λp/λα=√8.

Worked derivation:

$$p_p^2 = 2meV, \quad p_\alpha^2 = 2(4m)(2e)V,$$
$$\lambda_p/\lambda_\alpha = p_\alpha/p_p = \sqrt{8} = 2\sqrt{2}.$$

Why: Equal potential does not mean equal kinetic energy for unequal charges. Kinetic gain is qV, not simply V. The alpha's doubled charge gives $2eV$, while its fourfold mass controls speed through $K = \frac{1}{2}mv^2$; compare both before taking the speed ratio.

Why not the others: (A) takes only mass ratio's square root; (C) takes only charge ratio; (D) omits the square root.

Key point: Include both mass and charge in accelerated-particle momentum.

Q21 / D Limiting reagent with consecutive yields

Answer: (D) 0.030 mol.

Given: A limits the first reaction; the two yields apply successively.

Formula/Principle: Use stoichiometric ratios before applying each fractional yield.

Step 1: Theoretical C=0.10 mol, isolated C=0.10×0.80=0.080 mol.

Step 2: Theoretical D=0.080/2=0.040 mol, isolated D=0.040×0.75=0.030 mol.

Worked derivation:

$$n_{C,\text{theory}} = \min(0.10, 0.16) = 0.10 \text{ mol},$$
$$n_{C,\text{isolated}} = 0.80(0.10) = 0.080 \text{ mol},$$
$$n_{D,\text{isolated}} = 0.75(0.080/2) = 0.030 \text{ mol}.$$

Why: The second stage can use only C actually isolated, not the first stage's hypothetical maximum. The first reaction's limiting reactant fixes theoretical C, then the 80% isolation factor fixes usable C. The second reaction consumes that actual amount according to its 2:1 coefficient, so overall yield cannot be applied only at the end.

Why not the others: (A) misses the 2:1 step; (B) omits second yield; (C) applies yield without the reaction ratio.

Key point: Track material through each stage separately.

Q22 / A Substituent effects on benzoic-acid strength

Answer: (A) p-Nitrobenzoic acid.

Why: Nitro withdraws electron density through inductive and resonance effects and helps stabilise the negatively charged conjugate base. At para, methoxy resonance donation tends to oppose this stabilisation. Acid strength follows how favourably deprotonation products are stabilised under comparable conditions. A para nitro group withdraws through both induction and resonance, stabilising the carboxylate relative to neutral acid. A methoxy group donates by resonance at para, tending to destabilise that conjugate base in the relevant comparison.

Why not the others: (B) Donation does not stabilise the anion in the claimed way. (C) Ring substituents transmit electronic effects. (D) The same acidic group can have different environments.

Key point: Compare conjugate-base stabilisation, with substituent position specified.

Q23 / C Alkene reconstruction from ozonolysis fragments

Answer: (C) Propene.

Why: Each carbonyl carbon was originally one carbon of the C=C bond. Rejoining the methanal carbon and ethanal carbon by a double bond, replacing their carbonyl oxygens, gives CH_2=CH–CH_3. The fragments account for all three carbons in propene. Reductive ozonolysis splits the C=C and puts one oxygen at each original alkene carbon. Reconnecting the two carbonyl carbons and removing the oxygens reconstructs the original carbon skeleton and substituent placement.

Why not the others: (A) Gives two methanals. (B) Gives two ethanals. (D) Gives methanal plus propanone rather than ethanal.

Key point: Reconstruct the two former alkene carbons from their separate carbonyl products.

Q24 / B Aryl versus alkyl substitution

Answer: (B) Aryl C–Br is resistant to ordinary SN2.

Why: Conjugation with the ring gives the C–Br bond partial double-bond character, and backside approach to an sp^2 aryl carbon is geometrically unfavourable. These features distinguish aryl from primary alkyl halides under the stipulated mild conditions. Ordinary SN2 requires attack at a tetrahedral carbon from behind the leaving group. Chlorobenzene binds Cl to sp^2 aryl carbon and has resonance-strengthened C–Cl character, so that mechanism is inaccessible under the mild comparison.

Why not the others: (A) The bond exists. (C) Primary bromoalkane is not aromatic. (D) Stable free phenyl cations do not form readily in water.

Key point: Do not transfer an alkyl-halide mechanism uncritically to an aryl halide.

Q25 / D Phenol acidity relative to ethanol

Answer: (D) Phenoxide stabilisation.

Why: Deprotonating phenol gives an anion whose charge can delocalise into the aromatic system. Ethoxide lacks that resonance stabilisation, so ethanol is a substantially weaker acid under the usual qualitative comparison. Phenoxide has additional resonance contributors that spread its negative charge into the aromatic ring. Ethoxide is confined largely to oxygen, making its formation by aqueous hydroxide far less favourable.

Why not the others: (A) Ethanol has an O–H group. (B) Phenol is not an aldehyde. (C) The relevant reaction is proton transfer at oxygen.

Key point: Compare conjugate bases to explain an acidity contrast.

Q26 / A Charge of hydrolysed amino acids above pI

Answer: (A) Both tend net negative.

Why: Above each isoelectric point, deprotonation shifts the balance toward anionic forms. Glycine and alanine side chains do not introduce opposite ordinary ionisation behaviour, and hydrolysis frees their amino and carboxyl groups from the peptide bond. At high pH the carboxyl groups are deprotonated and the ammonium groups tend to lose protons. Evaluate each free amino acid's charge rather than assuming the intact peptide retains the same termini.

Why not the others: (B) Positive net charge is favoured at lower pH. (C) The non-ionising side chains do not reverse signs. (D) Net charge changes with pH.

Key point: Evaluate the free products at the stated pH, not the intact peptide.

Q27 / C Common-ion solubility approximation

Answer: (C) 1.0×10^{-9} M.

Given: $[Cl^-]\approx0.10$ M from fully dissociated NaCl.

Formula/Principle: $K_{sp}=[Ag^+][Cl^-]$ for AgCl(s).

Step 1: $[Ag^+]=10^{-10}/0.10$.

Step 2: This equals 10^{-9} M, far below added chloride and consistent with the approximation.

Worked derivation:

$$[Ag^+] = K_{sp}/[Cl^-] = 10^{-10}/0.10 = 10^{-9}\ \text{M} \ll 0.10\ \text{M}.$$

Why: Added chloride suppresses silver-ion concentration through the common-ion equilibrium. Write $K_{sp} = [Ag^+][Cl^-]$. The added 0.10 M chloride dominates the tiny chloride released by AgCl, so $[Ag^+] \approx K_{sp}/0.10$; check the resulting value is much smaller than 0.10 M.

Why not the others: (A) Is pure-water solubility; (B) omits division by chloride; (D) reports the chloride concentration.

Key point: Check the approximation after calculation.

Q28 / B Positive deviation from Raoult law

Answer: (B) Weaker unlike interactions.

Why: Mixing replaces some A–A and B–B neighbours with A–B neighbours. When those unlike contacts are less favourable on average, molecules escape more readily than the ideal reference predicts, raising total vapour pressure. Weaker unlike attractions make molecules escape more readily from the liquid than in the ideal mixture. A higher observed total vapour pressure is therefore a positive deviation; it does not imply that every component's mole fraction increased.

Why not the others: (A) Stronger unlike attractions generally cause negative deviation. (C) Contradicts pressure above a positive ideal prediction. (D) Positive deviation need not mean zero attraction.

Key point: Interpret vapour-pressure deviations through unlike contacts.

Q29 / D Copper sulfate electrolysis with inert electrodes

Answer: (D) Copper and oxygen.

Why: Cu^{2+} gains electrons at the cathode to form solid Cu. The stipulated anodic water oxidation releases O_2; sulfate remains in solution and does not become a gas. The electrode reactions must be assigned separately. At inert electrodes, the cathode and anode host separate reduction and oxidation half-reactions. Electrons consumed by Cu^{2+} deposition are balanced by electrons released from water oxidation under the specified conditions.

Why not the others: (A) Ignores the specified copper deposition. (B) Sulfate reduction is not the anode reaction. (C) Gaseous sulfate is not an ordinary product.

Key point: Identify reduction and oxidation at their respective electrodes.

Q30 / A Neutralisation calorimetry with limiting reagent

Answer: (A) −100.8 kJ/mol.

Given: Acid 0.0500 mol, base 0.0250 mol, so base limits and water formed is 0.0250 mol.

Formula/Principle: $q_{solution}=mc\Delta T$, $q_{reaction}=-q_{solution}$.

Step 1: $q_{solution}=100\times4.20\times6.0=2520$ J.

Step 2: $\Delta H=-2.520$ kJ/0.0250 mol$=-100.8$ kJ/mol water.

Worked derivation:

$$\begin{aligned} q_{\text{solution}} &= 100(4.20)(6.0) = 2520\ \text{J}, \\ n_{H_2O} &= \min(0.0500, 0.0250) = 0.0250\ \text{mol}, \\ \Delta H_m &= -2.520/0.0250 = -100.8\ \text{kJ mol}^{-1}. \end{aligned}$$

Why: The rising solution temperature means the reaction releases heat; dividing by the limiting amount fixes the molar basis. Equal solution mass and heat capacity turn the observed temperature rise into heat gained by solution. The reaction releases the opposite heat; divide by the actual limiting moles to report molar enthalpy with a negative sign.

Why not the others: (B) divides by all acid; (C) reverses heat sign; (D) halves reacting moles again.

Key point: Determine limiting product moles before normalising calorimetric heat.

Mathematics

Q31 / D Tangent-line parameter for an off-origin circle

Answer: (D) 0 and 50.

Given: Completing squares gives centre (3,4) and radius 5.

Formula/Principle: Tangency means centre-to-line distance equals radius.

Step 1: |3·3+4·4−k|/√(3²+4²)=5, hence |25−k|=25.

Step 2: k=0 or 50.

Worked derivation:

$$(x-3)^2+(y-4)^2=25, \quad \frac{|25-k|}{\sqrt{3^2+4^2}}=5 \Longrightarrow |25-k|=25.$$

Why: The two parallel tangents lie on opposite sides of the circle, yielding both signs of the absolute-value equation. Both signs of the absolute value must be retained because the two parallel tangent lines touch opposite sides of the circle.

Why not the others: (A) centres the circle at origin; (B) omits the radius-normal factor; (C) mistakes the centre's line value for a tangent.

Key point: Translate to centre–radius form before using point-to-line distance.

Q32 / C Implicit tangent and intercept

Answer: (C) x=2.

Given: Differentiating yields 2x+y+(x+2y)y′=0.

Formula/Principle: Implicit differentiation gives slope y′=−(2x+y)/(x+2y).

Step 1: At (1,1), slope=−3/3=−1.

Step 2: Tangent y−1=−(x−1), or x+y=2; set y=0 for x=2.

Worked derivation:

$$2x+y+(x+2y)y'=0, \quad y'(1,1)=-1; \qquad y-1=-(x-1) \Longrightarrow x+y=2.$$

Why: The product xy contributes both y and xy′; dropping one term changes the intercept. The tangent is determined by both its slope and the given point. Setting y=0 only after writing the full line avoids mistaking the point's x-coordinate for the intercept.

Why not the others: (A) sign error in line; (B) reuses point coordinate; (D) ignores nonzero slope.

Key point: Derive slope first, then use the point to form the tangent.

Q33 / B Open-box volume optimisation with a feasible domain

Answer: (B) (10−2√7)/3 cm.

Given: Height x, base dimensions 12−2x and 8−2x, with feasible 0<x<4 cm.

Formula/Principle: V=x(12−2x)(8−2x)=4x(6−x)(4−x); maximise inside the physical interval.

Step 1: V′=4(24−20x+3x²). Setting it to zero gives x=(10±2√7)/3.

Step 2: The plus root exceeds 4 cm and is infeasible; the minus root lies inside (0,4), where V rises then falls. Endpoint volume tends to zero.

Worked derivation:

$$V(x) = x(12-2x)(8-2x), \quad V'(x) = 4(3x^2 - 20x + 24) = 0, \quad x = \frac{10 \pm 2\sqrt{7}}{3}.$$

Why: A stationary point outside the domain cannot describe a box, even if it solves the algebraic derivative equation. The feasible interval is 0<x<4. The larger root violates it, while V vanishes at both limiting endpoints and the derivative changes from positive to negative at the smaller root.

Why not the others: (A) is the invalid derivative root; (C) is feasible but not stationary; (D) collapses a base dimension to zero.

Key point: Optimisation needs both calculus and feasibility checks.

Q34 / A Difference of logarithmic definite integrals

Answer: (A) (3/2)ln2−1.

Given: The two terms have different denominators and antiderivatives.

Formula/Principle: Substitute u=1+x² for the first; write x/(1+x)=1−1/(1+x) for the second.

Step 1: First integral=(1/2)ln2.

Step 2: Second integral=[x−ln(1+x)]₀¹=1−ln2.

Step 3: Difference=(3/2)ln2−1.

Worked derivation:

$$\int_0^1 \frac{x\,dx}{1+x^2} = \tfrac{1}{2}\ln 2, \quad \int_0^1 \frac{x\,dx}{1+x} = 1 - \ln 2, \quad I = \tfrac{3}{2}\ln 2 - 1 > 0.$$

Why: On 0<x<1, 1+x²<1+x, so the integrand is positive; the result is approximately +0.040, an independent sign check. The two antiderivatives require different substitutions. A numerical sign check agrees with the positive integrand on the interior, guarding against reversing the subtraction.

Why not the others: (B) reverses subtraction; (C) omits the second integral; (D) omits the substitution factor.

Key point: Integrate the two rational terms separately, then compare signs.

Q35 / D Domain of a composition with root and denominator

Answer: (D) x≥1 except 5.

Worked derivation:

$$\text{dom}(f) = [1, \infty), \quad g(f(x)) = \frac{1}{\sqrt{x-1} - 2}; \qquad \sqrt{x-1} = 2 \iff x = 5.$$

Why: The square root first requires x−1≥0. The outer function additionally forbids its input t=2; √(x−1)=2 occurs at x=5, which must be removed. Both conditions are needed to define the composition. The inner square root must be real before g can be evaluated. The outer denominator then removes a single otherwise valid point, not a whole interval.

Why not the others: (A) excludes valid x=1 and includes forbidden 5. (B) excludes valid 1≤x<5 and includes 5. (C) includes x<1 where the root is not real.

Key point: Apply inner-domain and outer-input constraints in order.

Q36 / C Scalar triple product with prescribed volume

Answer: (C) −1 and 5.

Given: Volume is |a·(b×c)|.

Formula/Principle: b×c=(−1,t−1,1), hence a·(b×c)=t−2.

Step 1: Set |t−2|=3.

Step 2: Solve both signs: t−2=±3, giving t=5 or −1.

Worked derivation:

$$\mathbf{b}\times\mathbf{c}=(-1,t-1,1),\quad \mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})=t-2,\quad |t-2|=3\Longrightarrow t=-1,5.$$

Why: Volume loses orientation sign, so two distinct parameter values can create the same magnitude. Volume is an unsigned magnitude, so opposite orientations of the three vectors can have the same positive volume.

Why not the others: (A) Gives zero volume; (B) uses an incorrect centre for the absolute value; (D) misses the negative branch.

Key point: Do not discard either sign of an absolute scalar triple product.

Q37 / B Determinant under coupled row combinations

Answer: (B) −6.

Given: Before interchange, the row-combination matrix is T with rows (1,1,0),(0,1,1),(1,0,1).

Formula/Principle: det(TA)=det T·det A; one row interchange changes sign.

Step 1: Expand det T=1(1)−1(−1)=2, so intermediate determinant=2×3=6.

Step 2: Interchange first and third rows to obtain −6.

Worked derivation:

$$\det T=\det\begin{pmatrix}1&1&0\\0&1&1\\1&0&1\end{pmatrix}=2,\quad \det(TA)=2(3)=6,\quad \det B=-6.$$

Why: These are three coupled row combinations, not three independent row additions that each preserve determinant. The simultaneous row combinations form a left multiplication with determinant two. The subsequent single row interchange changes sign once, independent of the row values.

Why not the others: (A) omits the final swap; (C) omits the transformation's factor 2; (D) omits both changes.

Key point: Represent simultaneous row combinations by a matrix before applying a later swap.

Q38 / A Conditional hypergeometric sample

Answer: (A) 2/11.

Given: Three red draws=C(4,3)=4; exactly two red=C(4,2)C(3,1)=18.

Formula/Principle: Conditional probability is favourable count divided by count satisfying the condition.

Step 1: At least two red includes 18+4=22 selections.

Step 2: All three red probability=4/22=2/11.

Worked derivation:

$$P(3R\,|\geq 2R)=\frac{\binom{4}{3}}{\binom{4}{2}\binom{3}{1}+\binom{4}{3}}=\frac{4}{18+4}=\frac{2}{11}.$$

Why: After conditioning, the denominator is not all C(7,3)=35 possible samples. Conditioning discards all outcomes with fewer than two red balls. The numerator and denominator must count unordered three-ball samples in the same way.

Why not the others: (B) is unconditioned probability; (C) treats two event types as equally likely; (D) uses an incorrect favourable count.

Key point: Build the conditioned sample space before dividing.

Q39 / D Lattice paths avoiding a point

Answer: (D) 66.

Given: All paths use five right and four up steps.

Formula/Principle: Count all sequences then subtract those visiting the forbidden point.

Step 1: Total=C(9,4)=126. Paths to (2,2): C(4,2)=6; paths onward: C(5,2)=10.

Step 2: Through-point paths=6×10=60; avoiding=126−60=66.

Worked derivation:

$$\binom{9}{4} - \binom{4}{2}\binom{5}{2} = 126 - (6)(10) = 66.$$

Why: The two subpaths can combine independently, which calls for multiplication before subtraction. A path through (2,2) decomposes uniquely into a path to that point and one from it to (5,4). These choices multiply; forbidden paths are then subtracted from all paths.

Why not the others: (A) ignores exclusion; (B) counts forbidden paths; (C) subtracts incorrectly.

Key point: Decompose paths at a required point and use complement for avoidance.

Q40 / C Vanishing coefficient followed by adjacent term

Answer: (C) −5.

Given: Multiplication by 1−x makes coefficient of x^r equal C(n,r)−C(n,r−1).

Formula/Principle: Set C(n,3)−C(n,2)=0; ratio C(n,3)/C(n,2)=(n−2)/3.

Step 1: (n−2)/3=1 gives n=5.

Step 2: x^4 coefficient=C(5,4)−C(5,3)=5−10=−5.

Worked derivation:

$$\binom{n}{3} = \binom{n}{2} \Longrightarrow \frac{n-2}{3} = 1 \Longrightarrow n = 5; \qquad [x^4](1+x)^5(1-x) = \binom{5}{4} - \binom{5}{3} = -5.$$

Why: The vanishing x^3 coefficient fixes the unknown exponent before the requested next coefficient can be evaluated. The vanishing coefficient determines n first. Only then can adjacent contributions to x^4 be evaluated, with the minus sign from the factor (1−x) retained.

Why not the others: (A) reverses subtraction; (B) retains only the shifted term; (D) assumes the next coefficient also vanishes.

Key point: Convolution with a short factor relates adjacent binomial coefficients.

Q41 / B Finite rational telescoping

Answer: (B) 17/30.

Given: Four terms have denominators 3,8,15,24.

Formula/Principle: 1/[k(k+2)]=(1/2)(1/k−1/(k+2)).

Step 1: Telescoping leaves (1/2)(1+1/2−1/5−1/6).

Step 2: The bracket is 17/15, giving 17/30.

Worked derivation:

$$\frac{1}{k(k+2)} = \frac{1}{2}\left(\frac{1}{k} - \frac{1}{k+2}\right), \quad S_4 = \tfrac{1}{2}(1 + \tfrac{1}{2} - \tfrac{1}{5} - \tfrac{1}{6}) = \frac{17}{30}.$$

Why: Interior reciprocal terms cancel after shifting by two. Writing out the first and last unmatched terms makes the shift-two telescoping visible and prevents the common error of cancelling too many boundary terms.

Why not the others: (A) stops at k=2; (C) stops at k=3; (D) accidentally sums 1/[k(k+1)] instead of the stated denominator.

Key point: Keep both unmatched terms at each end.

Q42 / A Root signs plus a threshold

Answer: (A) 0<k<2.

Given: Root product k−2; polynomial value at x=2 is −k.

Formula/Principle: Opposite root signs require product<0. With an upward-opening parabola and negative root<0, x=2 lies between the roots precisely when p(2)<0.

Step 1: k−2<0 gives k<2.

Step 2: −k<0 gives k>0, ensuring positive root>2. Combine to 0<k<2.

Worked derivation:

$$r_1 r_2 = k - 2 < 0, \quad p(2) = 2^2 - 2(k+1) + (k-2) = -k < 0 \Longrightarrow 0 < k < 2.$$

Why: At k=0 the positive root equals 2, while k=2 gives a zero root; both strict endpoints are excluded. For a monic quadratic with opposite-sign roots, p(2)<0 means 2 lies strictly between them and therefore below the positive root; equality at either endpoint fails the strict condition.

Why not the others: (B) includes values whose positive root is below 2; (C) includes forbidden endpoints; (D) loses opposite signs.

Key point: Combine Vieta's sign test with a polynomial-value test at the threshold.

Q43 / D Population variance from mean and sum of squares

Answer: (D) 3.

Given: $n = 5,\ \bar{x} = 4,\ \sum x_i^2 = 125.$

Formula/Principle: Population variance is the mean square minus the square of the mean.

Step 1:

$$\sigma^2 = \frac{125}{5} - 4^2 = 25 - 16 = 9.$$

Step 2:

$$\sigma = \sqrt{9} = 3.$$

Worked derivation:

$$\sigma = \sqrt{\frac{1}{n}\sum x_i^2 - \bar{x}^2} = 3.$$

Why: The observations constitute the stated population, so the divisor is five rather than four. Standard deviation is the square root of variance and has the same units as the observations.

Why not the others: (A) reports variance; (B) reports the total squared deviations; (C) takes the square root before averaging those deviations.

Key point: Distinguish a squared-deviation total, variance and standard deviation.

Q44 / C Eliminating constants from a solution family

Answer: (C) y″=y.

Worked derivation:

$$y' = C_1 e^{x} - C_2 e^{-x}, \qquad y'' = C_1 e^{x} + C_2 e^{-x} = y.$$

Why: Differentiating once gives y′=C₁e^x−C₂e^(−x), so a first-order equation y′=y cannot accommodate the second term. Differentiating again restores the plus sign: y″=C₁e^x+C₂e^(−x)=y for all constants. Two independent constants remain free, so checking only a special choice such as C₂=0 would not establish an equation for the entire family.

Why not the others: (A) fails whenever $C_2 \neq 0$. (B) reverses the second-derivative sign. (D) holds only for the zero function in this family.

Key point: Differentiate enough times to eliminate independent arbitrary constants.

Q45 / B Continuous extension and derivative at a point

Answer: (B) k=2 and derivative zero.

Given: $\sin(2x)/(2x) \to 1$, hence $f(x) \to 2$.

Formula/Principle: Continuity fixes k=2; derivative is $\lim_{x\to 0}(\sin 2x/x - 2)/x$.

Step 1: Using $\sin 2x = 2x - (2x)^3/6 + o(x^3)$, the numerator is $-(4/3)x^2 + o(x^2)$.

Step 2: Divide by x to obtain $f'(0) = 0$.

Worked derivation:

$$\sin(2x) = 2x - \frac{(2x)^3}{6} + O(x^5), \quad f(x) = 2 - \frac{4}{3}x^2 + O(x^4), \quad k = 2,\ f'(0) = \lim_{x\to 0}\frac{f(x)-2}{x} = 0.$$

Why: A finite continuous value does not itself determine the derivative; a second limiting step is required. The continuous extension sets the value at zero; differentiability requires a separate difference quotient after cancellation, not just a repeated use of the first sine limit.

Why not the others: (A) fails continuity; (C) confuses limit with derivative; (D) ignores the standard sine limit.

Key point: Extend continuously before differentiating at a removable point.

Physics

Q46 / A Independent vertical motion of projectiles

Answer: (A) Simultaneously.

Worked derivation:

$$y_1(t) = h - \tfrac{1}{2}gt^2 = y_2(t), \qquad t_{\text{hit}} = \sqrt{2h/g}.$$

Why: Both start with zero vertical velocity and accelerate downward at g from the same height. Their vertical displacements are identical functions of time, while horizontal velocity changes only where the first ball lands. The horizontal launch changes x(t) but not the initial vertical velocity or vertical acceleration. This assumes air resistance is negligible and both are released at the same instant.

Why not the others: (B) and (C) incorrectly let horizontal motion alter fall time. (D) Horizontal speed affects range but not vertical flight time in this model.

Key point: Resolve projectile motion into independent components.

Q47 / D Instantaneous power as a dot product

Answer: (D) 2 W.

Given: Both force and velocity are vectors with x and y components.

Formula/Principle: $P = F \cdot v$, the rate of work along instantaneous motion.

Step 1: Multiply matching components: $2\times 3 + (-1)\times 4 = 6 - 4$.

Step 2: $P = 2$ N·m/s $= 2$ W.

Worked derivation:

$$P = \mathbf{F}\cdot\mathbf{v} = (2)(3) + (-1)(4) = 2\,\text{N m s}^{-1} = 2\,\text{W}.$$

Why: The force's negative y component removes some of the power added by its x component. The dot product extracts force along instantaneous velocity; the negative y contribution cancels part of the positive x contribution.

Why not the others: (A) adds absolute contributions; (B) uses speed alone; (C) reverses the dot-product sign.

Key point: Power depends on the component of force along velocity.

Q48 / C Eccentric cutout moment of inertia

Answer: (C) 13MR²/32.

Given: Uniform surface density makes removed mass M/4. The figure gives removed radius R/2 and displacement d=R/2.

Formula/Principle: Subtract the hole's inertia about O using the parallel-axis theorem.

Step 1: Original I=MR²/2; hole's central I=(M/4)(R/2)²/2=MR²/32.

Step 2: Hole's displacement term=(M/4)(R/2)²=MR²/16.

Step 3: Remaining I=MR²(1/2−1/32−1/16)=13MR²/32.

Worked derivation:

$$m_{\text{hole}} = M/4, \quad I_O^{\text{hole}} = \tfrac{1}{2}(M/4)(R/2)^2 + (M/4)(R/2)^2 = \frac{3MR^2}{32}, \quad I_{\text{remain}} = \frac{MR^2}{2} - I_O^{\text{hole}} = \frac{13MR^2}{32}.$$

Why: The missing circle is not concentric with O, so subtracting only its central inertia is insufficient. The removed circle contributes its own central inertia plus a parallel-axis shift about O. Using the final remaining mass M in the full-disc formula would change the meaning of M in the stem.

Why not the others: (A) mishandles removed mass; (B) omits part of parallel-axis shift; (D) treats the hole as central.

Key point: Treat removed material as a negative body at the same axis.

Q49 / B Free-surface slope in an accelerating tank

Answer: (B) 1/2.

Given: Effective gravity is g down plus g/2 left.

Formula/Principle: A free surface is perpendicular to effective gravity; |Δh/Δx|=a/g.

Step 1: Substitute a=g/2.

Step 2: Slope magnitude=(g/2)/g=1/2; it is higher on the left.

Worked derivation:

$$\mathbf{g}_{\text{eff}} = -(g/2)\hat{\mathbf{x}} - g\hat{\mathbf{y}}, \qquad \left|\frac{dh}{dx}\right| = \frac{g/2}{g} = \frac{1}{2}.$$

Why: In the accelerating frame, the horizontal pseudo acceleration tilts the effective gravity vector. The free surface is perpendicular to effective gravity. The horizontal fictitious acceleration points left in the tank frame, making the fluid higher on that side.

Why not the others: (A) inverts a/g; (C) assumes a=g; (D) squares the ratio.

Key point: Separate the slope magnitude from its left-right direction.

Q50 / A Entropy of thermal equilibration

Answer: (A) C ln(4/3).

Given: Energy balance gives final temperature (T+3T)/2=2T.

Formula/Principle: For each body, ΔS=C ln(Tfinal/Tinitial).

Step 1: Add C ln(2T/T)+C ln(2T/3T).

Step 2: Combine to C ln(4/3)>0.

Worked derivation:

$$2CT_f = C(T + 3T) \Longrightarrow T_f = 2T; \quad \Delta S = C\ln\frac{2T}{T} + C\ln\frac{2T}{3T} = C\ln\frac{4}{3}.$$

Why: Equal and opposite heat transfer between bodies still creates net entropy when it occurs

across a finite temperature difference. Both bodies share one isolated final temperature from energy balance. Entropy changes must be calculated separately for hot and cold bodies before adding; zero net heat does not mean zero entropy production.

Why not the others: (B) confuses zero net heat with zero entropy; (C) reverses the logarithm; (D) ignores the hot body's entropy loss.

Key point: Sum each body's state-function change.

Q51 / D Spatial phase difference

Answer: (D) π/3.

Given: One full wavelength corresponds to 2π spatial phase.

Formula/Principle: Δφ=2πΔx/λ.

Step 1: Substitute Δx=λ/6.

Step 2: Δφ=2π/6=π/3.

Worked derivation:

$$\phi(x,t) = kx - \omega t + \phi_0, \quad |\Delta\phi| = k\Delta x = \frac{2\pi}{\lambda}\frac{\lambda}{6} = \frac{\pi}{3}.$$

Why: Comparing positions at the same instant isolates the spatial phase term rather than time evolution. Comparing two positions at the same instant cancels the time-dependent term. The result is a phase difference, not a path length or a time delay.

Why not the others: (A) omits the factor two; (B) uses a quarter wavelength; (C) doubles the result.

Key point: Convert displacement to phase as a fraction of a full cycle.

Q52 / C Total energy of a circular satellite orbit

Answer: (C) −GMm/(2r).

Given: The circular-orbit condition gives mv²/r=GMm/r².

Formula/Principle: K=mv²/2=GMm/(2r), U=−GMm/r.

Step 1: Add K and U.

Step 2: E=GMm/(2r)−GMm/r=−GMm/(2r).

Worked derivation:

$$\frac{mv^2}{r} = \frac{GMm}{r^2} \Longrightarrow K = \frac{1}{2}mv^2 = \frac{GMm}{2r}; \quad E = K + U = \frac{GMm}{2r} - \frac{GMm}{r} = -\frac{GMm}{2r}.$$

Why: The negative result signifies a bound orbit relative to zero energy at infinity. Circular orbit fixes speed through gravity; gravitational potential energy remains negative with zero chosen at infinity. The total bound energy is half the potential energy, not the same quantity.

Why not the others: (A) loses binding sign; (B) reports U alone; (D) falsely equates kinetic and potential magnitudes.

Key point: Total energy is not the same as gravitational potential energy.

Q53 / B Zero potential on a signed line

Answer: (B) −a and 3a/5.

Given: The signed-axis diagram fixes charge positions and two sign-change intervals.

Formula/Principle: V=kq/|x|−4kq/|x−3a|=0, hence |x−3a|=4|x|.

Step 1: For x<0, 3a−x=−4x gives x=−a.

Step 2: For 0<x<3a, 3a−x=4x gives x=3a/5; for x>3a, the formal solution fails that interval.

Worked derivation:

$$\frac{1}{|x|} = \frac{4}{|x-3a|} \Longleftrightarrow |x-3a| = 4|x|; \quad x<0:\ x=-a, \quad 0<x<3a:\ x=3a/5.$$

Why: Potential is scalar but distances remain positive, so absolute values need separate intervals. Distances must remain nonnegative in each region, and neither charge location is a valid finite-potential point. A formal root outside the interval assumed to derive it must be rejected.

Why not the others: (A) uses equal charge magnitudes; (C) names singular charge positions; (D) chooses distances with incorrect interval signs.

Key point: Verify every algebraic solution in its positional interval.

Q54 / A Plate separation change at fixed battery voltage

Answer: (A) Both become half.

Worked derivation:

$$C_i = \frac{\varepsilon A}{d}, \quad C_f = \frac{\varepsilon A}{2d} = \frac{C_i}{2}; \qquad Q_f = C_f V = \frac{Q_i}{2}, \quad U_f = \tfrac{1}{2} C_f V^2 = \frac{U_i}{2}.$$

Why: Capacitance εA/d halves when d doubles. The connected ideal battery holds V fixed, so Q=CV and U=CV²/2 each scale with C and halve. The battery can accept the charge that leaves the capacitor. The connected ideal battery fixes voltage but permits charge exchange. Applying Q constant would describe a disconnected capacitor and produce a different energy trend.

Why not the others: (B) Reverses capacitance change. (C) Assigns different scaling to quantities both proportional to C at fixed V. (D) Treats an attached battery as if it were disconnected.

Key point: Identify the fixed voltage boundary condition before applying capacitance formulas.

Q55 / D Terminal voltage with internal resistance

Answer: (D) 8 V.

Given: Total series resistance 6 Ω.

Formula/Principle: I=E/(R+r), Vterminal=IR=E−Ir.

Step 1: I=12/6=2 A.

Step 2: Terminal voltage=2×4=8 V.

Worked derivation:

$$I = \frac{\mathcal{E}}{R + r} = \frac{12}{4 + 2} = 2\,\text{A}; \qquad V_{\text{terminal}} = IR = (2)(4) = 8\,\text{V} = \mathcal{E} - Ir.$$

Why: Two volts of internal resistance at 2 A account for 4 V dropped inside the cell. The internal resistor dissipates Ir=4 V at this current, so the external terminal voltage is below the open-circuit emf.

Why not the others: (A) assumes no internal drop; (B) reports that drop; (C) treats resistance sum as voltage.

Key point: Distinguish emf from voltage under load.

Q56 / C Torque on a current loop

Answer: (C) (√3/2)IAB.

Given: Loop normal makes 60° with the field.

Formula/Principle: Torque magnitude=IAB sinθ, where θ is normal–field angle.

Step 1: θ=90°−30°=60°.

Step 2: τ=IAB sin60°=(√3/2)IAB.

Worked derivation:

$$\theta_{\mu B} = 90^\circ - 30^\circ = 60^\circ, \qquad \tau = \mu B \sin\theta_{\mu B} = IAB \sin 60^\circ = \frac{\sqrt{3}}{2} IAB.$$

Why: The magnetic moment is normal to the loop's plane, so the stated plane angle is not directly θ. Torque depends on the angle between the field and the loop's normal, which is the magnetic moment direction, not the given plane–field angle.

Why not the others: (A) substitutes 30°; (B) assumes 90°; (D) assumes alignment of normal and field.

Key point: Convert a plane angle to the dipole-axis angle.

Q57 / B Ideal transformer voltage and power

Answer: (B) 60 V and 2.5 A.

Given: Turns ratio Ns/Np=1/4 and secondary resistor 6 Ω.

Formula/Principle: Vs/Vp=Ns/Np; ideal input power equals output power.

Step 1: Vs=240/4=60 V; Is=Vs/R=10 A.

Step 2: VpIp=VsIs=600 W, so Ip=600/240=2.5 A.

Worked derivation:

$$V_s = 240\frac{50}{200} = 60\,\text{V}, \quad I_s = \frac{60}{6} = 10\,\text{A}, \quad I_p = \frac{V_s I_s}{V_p} = \frac{600}{240} = 2.5\,\text{A}.$$

Why: Lower voltage accompanies higher secondary current for conserved power. Ideal power conservation connects currents on different windings; the load current is not the same as the supply current of a step-down transformer.

Why not the others: (A) reports secondary current as primary; (C) ignores turns ratio; (D) uses a half instead of quarter ratio.

Key point: Combine turns ratio with load and power conservation.

Q58 / A Optical path excess through a slab

Answer: (A) (n−1)L.

Given: Slab path length L and refractive index n.

Formula/Principle: Optical path is n times geometric distance in that medium.

Step 1: Slab optical path=nL; vacuum path=L.

Step 2: Excess=nL−L=(n−1)L.

Worked derivation:

$$\text{OPL}_{\text{slab}} = nL, \quad \text{OPL}_{\text{vacuum}} = L, \qquad \Delta\text{OPL} = nL - L = (n-1)L.$$

Why: The question asks for a comparison, so total slab optical path alone is not the final answer. The excess compares two paths of the same geometric length. Referencing the vacuum path avoids mistaking the slab's total optical length for the requested difference.

Why not the others: (B) is total path; (C) is not the defined optical path; (D) adds instead of subtracts the reference.

Key point: Subtract the specified baseline from optical path.

Q59 / D Propagation from E and B orientation

Answer: (D) +x.

Worked derivation:

$$\mathbf{S} \parallel \mathbf{E} \times \mathbf{B}, \qquad \hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}.$$

Why: Energy transport and propagation follow the Poynting direction E×B. For a right-handed coordinate system, ŷ×ẑ=x̂, perpendicular to both fields. The Poynting vector defines energy-flow direction in a plane wave; reversing cross-product order would reverse propagation, and propagation is perpendicular to both fields.

Why not the others: (A) Reverses cross-product order. (B) and (C) put propagation along one field, whereas a plane electromagnetic wave is transverse.

Key point: Use field orientation, not field magnitude, for propagation direction.

Q60 / C Two half-lives: activity and surviving fraction

Answer: (C) 1:2 and 5/32.

Given: X has undergone four half-lives, leaving $N_0/16$; Y two, leaving $N_0/4$.

Formula/Principle: Activity $A=\lambda N$ with $\lambda=\ln 2/T_{1/2}$; total survivor fraction $=(N_X+N_Y)/(2N_0)$.

Step 1: $\lambda_X/\lambda_Y=2$, so $A_X/A_Y=2(1/16)/(1/4)=1/2$.

Step 2: Surviving fraction $=(1/16+1/4)/2=5/32$.

Worked derivation:

$$\frac{N_X}{N_0}=2^{-8/2}=\frac{1}{16},\quad \frac{N_Y}{N_0}=2^{-8/4}=\frac{1}{4};\quad \frac{A_X}{A_Y}=\frac{\lambda_X N_X}{\lambda_Y N_Y}=\frac{2(1/16)}{1/4}=\frac{1}{2},\quad \frac{N_X+N_Y}{2N_0}=\frac{5}{32}.$$

Why: More X nuclei have decayed, but its decay constant is twice as large, so population ratio alone is not activity ratio. Activity depends on both survivor count and decay constant. The two initial populations contribute equally to the denominator of the overall survival fraction.

Why not the others: (A) uses population ratio as activity; (B) reverses it; (D) reports the decayed fraction.

Key point: Keep population and activity distinct for each isotope.